

HR Analytics
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Lecture 04: Pitfalls of data science framework and Basics of data visualization

Dear participants, in this session, we will learn some of the weaknesses of the data science framework and basics of data visualization. If we have to visualize the data, so what are the things that we should remember before visualizing the data, how we should visualize, so what are the things that we should consider and some of the weakness of the data science framework. So, these are the two things that we will discuss in this session. Right, pitfalls of the data science framework and basics of data visualization. So, first weakness that is what you can say confirmation trap, right. So, this is the trap that most people may face during the data analysis, right.

So, what happens when we pay more attention to our belief right and ignore the facts and patterns in the data, then we fall into the confirmation bias, right. I hope you understand what I am trying to make you understand. Like somebody who is an experienced person, right, so that person might have come across with so many findings, so because of that that person's belief got developed, this should happen in this way, this is how it should happen. So because of that, sometimes we ignore the facts and patterns that are there in the data, but we are not able to see, or we do not pay much attention to them.

So, if this is the indication, then that person may fall into the confirmation trap. That is what may happen with the, I am not saying it happens with everyone, but it may happen with some of the people. Similarly, I can tell you one principle: the things in which you believe more, you try to see or that are visible more in front of you. So, that is the belief that you will see, because this is the impact of the belief. If you believe in something, that is how it happens, then your attention towards that particular thing will be more, and because of that, you may ignore some of the things which you should not do.

So, it may happen. So, you have to think before and after analysing the data, have you suffered from the confirmation trap or not? So, that is what you need to recheck your results; you need to reconfirm the results and check it again and again so that you confirm you are not suffering from the confirmation trap. That is what may happen with you while analysing the data. So, you need to be very, very careful about after the data

analysis about the confirmation trap. The second trap that you may face is the overconfidence trap, right? So, what happens in this case? Like senior decision-makers who have been promoted to higher positions in the organization based on past success like in the past whatever decisions they have made.

So, they have proved their decision-making abilities throughout their careers right. So, because of that, what happens is a person gets overconfident right and does not pay much attention to the question questioning methods, motivation, or communicating the findings. So, basically, I can say that when somebody gets huge experience while doing data analysis, that person sometimes does not give much attention to the research method process, that research process, identifying the problem, developing the hypothesis, and collecting the data. So, sometime it happens that this researcher may not, do not pay more attention to the entire process. So, because of that, because of that negligence, we miss some of the steps.

Or we do not go in-depth as much a priority we should give to each step that we do not give because of the overconfidence that our decision maker has; he may say, I have done this particular activity so many times. I have done this particular activity so many times, because of that I should not recheck it, right. So, there is such kind of mistakes that senior decision make, decision makers may make. I am not saying that always they make such kind of mistake, but sometime it happens they make. So, whenever you have analyzed the data and whenever you are conducting this data, research process, so the research process has to be in-depth, rigorous, so that the impact of overconfidence bias of some of the people is not there on the on the results.

Because of this overconfidence about the research process, each step of the research process may suffer, and because of that, you may not reach the right decision or more appropriate decision where you have reached that it may be possible. So, you need to recheck the entire process at any step. Have you suffered from overconfidence bias or not? The next step that you may be facing is the overfitting trap. Sometimes, it may be possible you are getting results which you have not expected and some surprising results that you are getting right. So, because of that, you are excited, and you are trying to fit that particular result, right? That is what you are trying to make that result fit.

So, that is also may happen during the research process whenever you are making this decision. So, now what do you need to do? You need to think about all three types of traps, whether it is overfitting, confirmation or overconfidence of the decision maker. Any of the traps you are in, then you will not be able to reach the right decision, and you will not be able to do an analysis of your data, whatever data you have at hand. So, while doing this data analysis, you have to be very, very cautious about this decision-making

process. And the fourth thing that I always say is do not let your analytics hide the truth, right? So what do you need to do in this case because when you are calculating this basically mean, mean value, right.

So that may be impacted by the outlier. So, you need to be very, very careful about the outliers in your data, right. For example, for example, average age of 5 employees, right. So, for example, average is of 5 employees, one employees age is 20, another one is 21, 23 and 20, 24 and one employees age is 80. So, because of this 80, one employee's age, this entire mean of these 5 employees may be impacted.

Whenever you are doing this analysis, you have to report this outlier, one of the employee that is having the, that employee is 80 year old. So, when you are highlighting, because if you will not highlight, then the average age of employee that is there, approximately 22, that is the approximate that age is there, average age is there. So, this age may be impacted because of this 80, one outlier is there and because of that you will not be able to make the right kind of projection about the average age of the employee. So, such kind of outlier whether it is related to the work experience, whether it is related to the salary, whether it is related to the age. it is related to number of application right from where they are coming.

So, any outlier is there in term of the source of application any outlier is there that you need to identify you have to report it. So, that the truth of the data will not be hidden. If you do not report the outlier, then the possibility is very, very high, and you will not be able to tell the truth that your data is telling. So, what is my suggestion to the all manager whenever you are analyzing the data please check your data whether it is having any outlier or not. If outliers are there then you have to be very very cautious about the data I hope it is clear to you right.

Now whenever you have to make the visualization. So, basics of the visualization if you understand. So, first thing that you need to understand know your audience. You should know your audience who is your audiences right. So, in this case, what can say see X and Y? That is what you can see.

So, see the two dimensions are is there one is X, another one is Y. On X you can see analytics expertise, on X you can see analytics expertise. So, whoever is your audience whether that audience is expert of analytics or no. So, low to high. So, you need to, so the one dimension that is measuring the analytics expertise.

Second dimension is measuring the knowledge of the specific analytics project. So, whatever project that you are presenting, so knowledge about that particular project. So,

how much knowledge that audience is having? So, that is what you need to see low to high. So, in this case, so that is how you can divide your all audience into the four category right. So, if knowledge about these knowledge of the specific analytics project is high and analytics expertise also very high then you can say they are the master.

So, presentation data visualization for these master has to be different from the newcomer because newcomer may not understand so many terms and that you have used in your presentation, right? Although these people are some of the people that category that we discussed first category that is the master, so they are the expert of the analytics as well as they are having the knowledge of the specific analytics project. So, you cannot give the same presentation to the all four categories. So, you need to understand, so what you can do before making a data visualization presentation in front of anyone, what you do, you make this 2 by 2 matrix, understand in which quadrant the audience is coming from, whether it is coming in the master, whether it is coming in the scientist, whether they are coming in the category of newcomer or they are the enthusiast. Analytics expertise is low, but knowledge about the project is very high. In the same way you can see low, expertise is low, knowledge also very low.

So, they are the newcomer, just they have started. So, presentation for this newcomer has to be different from the masters, masters presentation has to be different from the scientist and scientist presentation has to be different from the enthusiast. So, who is your audience that you need to understand and as per the expertise of that audience you have to develop your presentation or you have to visualize your data that is what you need to do it. So, that is this is the one of the way that I can suggest you to divide your audience into four categories and as per after dividing the this audience you can make your presentation. So, what are the things that you need to consider in a presentation we will discuss in coming slides.

So, now, what type of things can you ask like based on this four quadrant? So, first you need to understand who is your audience. I already said how to do that two by two matrix that scientist, newcomers, master. So, in which category your audience comes. So, answer that question. They are the master, they are the scientist, they are the newcomer.

How will it read it? And interpret the information. So, how newcomer will read it? How newcomer will interpret it? So, newcomer may not be aware about the each and every terms. So, what you need to do? You have to give a detail of each and n terms. What is x axis? What is y axis? How many numbers are there? What is the percentage? So, everything has to be in detail for the newcomer. Opposite to that for a master it is not necessary everything has to be in detail, but depth of the information that has to be more right.

So, depth that you can say coverage of that analysis interpretation that has to be more in the case of the master. So, that is what you can think of. So, the first question is what you can think of before visualizing it: who is your audience and how do they read and interpret it? The second thing that you can do is assume these individuals have knowledge of the terminology and concepts that you will see. So, I already said the moment you have assumed the category and then you have to think whether they have the understanding about the terminology and concepts that you are going to use or not. Next, you need to understand or So, do they need any assistance from the presenter, from that visualization to understand that visualization? If they need any assistance, you provide it.

So, in the case of newcomer and entourage, they may need little bit more information than the masters and scientist, right? So, that is what you need to understand. So, that is what the note that I have written is, and the audience of the expert will have a more expectation from the general audience. So, expectation, expert will expect you should be able to drive some meaningful interpretation from the visualization. So, they may be interested in more, they may be interested in the quality of information that you are presenting through the visualization.

So, visualization tools. The next thing that you need to understand is what the visualization's functional role is and how viewers can take action from it right. So, after visualizing you need to understand how viewers whoever is viewing that visualization how they can make a decision based on that, how they can understand, how they can use that particular information that you have visualized it. So, that is what you need to think about. So, these are the questions that you can think in detail before visualizing any data through the Excel by using Excel, Power BI or Tableau. So, next what is the aim of the data visualization that also you need to understand.

So, aim of the data visualization is to confirm the sum of the information. That is the purpose or is to give a, to educate someone, that is the objective or to explore the new insights and new things. So, what is the objective? So, if objective is confirmation, then your way of presentation has to be different. If your objective is to educate something, someone, then your presentation has to be different. If your objective is to explore something, some new thing that you want to explore, then then your presentation has to be different.

So, in all three category the presentation has to differ from one category to another category. So, before visualizing the data you need to understand the aim of the visualization. So, how you can understand by understanding your audience carefully. So, what this your audience is expecting? It is expecting to get it confirmed, or it is expecting

to get more educated about that particular data, or it is that the audience is having to want to explore a little bit more in-depth about the problem. So, what is the objective? That is what you need to understand the objective of the data visualization.

So, if you find the right objective and match it with the right visualization strategy, your audience will be happier, or you will be able to gain a clearer understanding of the data to your audience. So, after making the graphs, what do you have to do? So, but question comes how you should make the graph first and then what questions that you should ask. So, before making the graph, I always suggest how you should make the graph in coming sessions. also, you will listen when I will use Power BI, Excel or Tableau. In that also I will say clearly if you have understood this data science framework.

So, the first thing that I focused on was HR issues. So, the moment you have identified this HR issues, from these HR issues now you need to identify the variables, variables. So, like you have identified age, gender, work experience, expected salary, performance rating. So, for example, these are the variables that you have identified to understand one of the problems. So, what you can do here now in the moment you have in you have collected the data in and then you have transform it you have make it appropriate for the analysis right.

So till here you have reached and that is what you have done it. After this what you have to do you have to think if you will make a age and gender are the two variable. So, how you want their presentation? Age and work experience, how you want to present it? So, whether you want to make age and gender together, you want to present it along with work average work experience also you want it. So, in this case you can make it a table.

If you want to make it only age and gender. In this case you can see age is a continuous and gender is a categorical. So in this case you can make a bar graph, right, an average age of male employees and average age of female employees, right. The second thing that you can do, you can divide this age into categories, right, various categories that you can divide an age and then you can check how many males are there, and how many females are there in that particular category. So, what I am trying to make you understand, before visualizing this data, you understand the nature of variables, right, nature of variables. So, gender is categorical, work experience is continuous, expected salary is continuous, and performance rating is you can say interval.

So these are the categories that you need to understand and how these categories can be presented well. And when you have to average it, when you have to sum it, when you have to multiply right, that nobody is going to teach you, right, this basic mathematics understanding that you need to have by yourself. Nobody can make you understand in

this case you have to multiply, in this case you have to add it. So, what do you have to do? You have to have a basic understanding of the variable. So, understand the nature of the variable, how you want to present it and you have thought this is the graph that you have to make it, right.

Go to that particular tool and make that particular graph right. And before making the graph you understand through this graph what you want to communicate. So, ask question what message am I trying to convey through this particular graph. Ask yourself what message that you want to give it. So, if your message and your visualization both things are matched right then you can say that yes it is a right kind of visualization right.

The second question that you can ask is, Do my visuals accurately reflect the number? Whatever number that you have presented, so if you will see in coming sessions, I will show some of the graphs which are there, but participants forgot to put the number. So, whatever number that you want to put it, whether in the form of number, absolute number, that introduction session that if you remember, I had discussed the way of measuring the variable. So, how you want to measure it? You want to measure it in term of the percentage, you want to measure in term of the percentile, you want to measure in term of the percentage, percentile in term of the number. So, how you want to measure it? So, is it reflecting the same number in which you want to measure it? So, if that is what you want to do it, so you have to check your visuals accurately. Third, an important question that you need to ask yourself is whether my data is memorable or not, right? If through the visualization if some of the people are able to memorize your data, it means your visualization is very, very good.

If people are not able to visualize the data, not able to memorize the data, then you can say your data is not that good. So, can you visualize the data in such a way that people are able to remember your data, whatever data you are going to present in any report, or you are going to make a presentation in any PowerPoint presentation? So, people are able to memorize your graphs, they are able to memorize your tables, anything that if the answer is yes, then you can say your data visualization is very good, right? So, I hope you would have understood the process of understanding the audience and importance of the audience, right, in the data visualization. That is the one thing few principles related to data visualization, how you can make a better graph, what you should know before using this tool and techniques of data visualization, right? So, I hope you would have understood these basic things related to this and the weaknesses of the data science framework. So, thank you for understanding this session.